

[Aim: 100/100 in Maths]

आरंभ

ARITHMETIC PROGRESSION

Lecture #9

#LP: An A.P. consists of 37 terms. The sum of the three middle most terms is 225 and the sum of last three terms is 429. Find the A.P.

$$n = 37 \leftarrow \text{odd}$$
$$MT = \left(\frac{n+1}{2}\right)^{\text{th}}$$
$$= \left(\frac{37+1}{2}\right)^{\text{th}}$$
$$MT = \left(\frac{38}{2}\right)^{\text{th}}$$
$$MT = 19^{\text{th}}$$

$$a_{18}, a_{19}, a_{20}$$
$$\text{A.T.O. } a_{18} + a_{19} + a_{20} = 225$$
$$a + 17d + a + 18d + a + 19d = 225$$
$$3a + 54d = 225$$
$$\cancel{3(a + 18d)} = \cancel{225} \quad 75$$
$$a + 18d = 75 \quad \text{--- (I)}$$

$$\text{A.T.O. } a_{35} + a_{36} + a_{37} = 429$$
$$a + 34d + a + 35d + a + 36d = 429$$
$$3a + 105d = 429$$
$$\cancel{3(a + 35d)} = \cancel{429} \quad 143$$
$$a + 35d = 143 \quad \text{--- (II)}$$
$$a = 3$$
$$d = 4$$

AP

$$\underset{?}{a}, \underset{?}{a+d}, \underset{?}{a+2d}, a+3d - - -$$

$$, 3, 3+4, 3+2(4), 3+3(4)$$

~~Ex~~

$$AP \Rightarrow 3, 7, 11, 15 - -$$

#LP: In a school, students thought of planting trees in and around the school to reduce noise pollution and air pollution. It was decided that the number of trees that each section of each class will plant be the same as the class in which they are studying e.g. — a section of I class will plant 1 tree, a section of II class will plant 2 trees and so on a section of class XII will plant 12 trees. There are three sections of each class. How many trees will be planted by the students? [CBSE 2014]



total trees $\Rightarrow 3 \times 1 + 3 \times 2 + 3 \times 3 + \dots + 3 \times 12$

$\Rightarrow 3 [1 + 2 + 3 + \dots + 12]$

$\Rightarrow 3 \left[\frac{n}{2} (a + l) \right] \Rightarrow 3 \left[\frac{12}{2} (1 + 12) \right] \Rightarrow 3 [6 (13)]$

$\Rightarrow 3 \times 78 \Rightarrow \boxed{234 \text{ trees}}$

#LP: A spiral is made up of successive semi-circles, with centres alternately at A and B, starting with centre at A, of radii 0.5 cm, 1.0 cm, 1.5 cm, 2.0 cm, ... as shown in Fig. what is the total length of such a spiral made up of thirteen consecutive semi-circles? Take $\pi = 22/7$

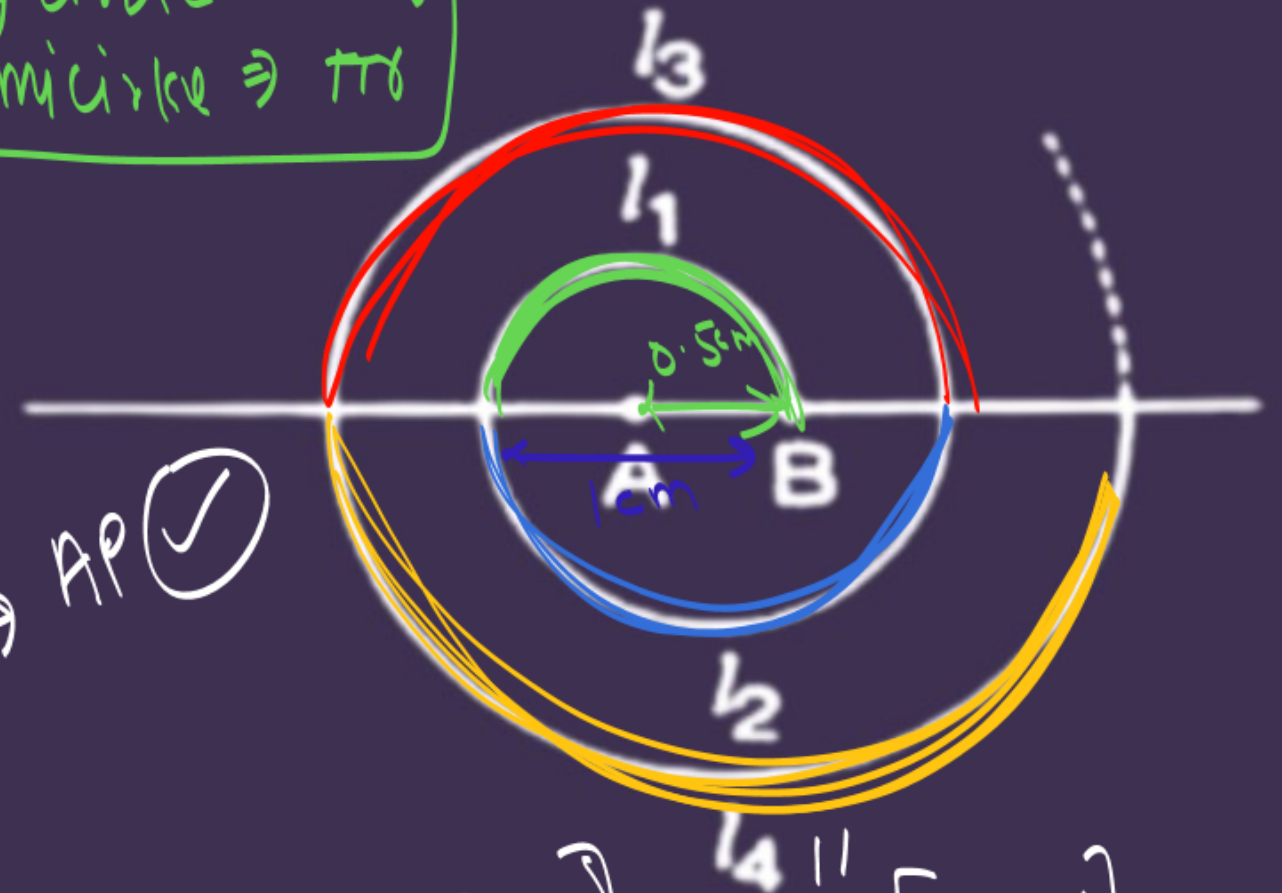
Circ. of circle = $2\pi r$
Semicircle $\Rightarrow \pi r$

$$\begin{array}{ccccccc} \pi(0.5) & , & \pi(1) & , & \pi(1.5) & , & \pi(2) & - & - & - \\ l_1 & & l_2 & & l_3 & & l_4 & - & - & - \end{array}$$

AP $\Rightarrow \pi (0.5 + 1 + 1.5 + 2 + 2.5 - - -)$ $\rightarrow$ AP $\checkmark$

$$S_{13} \rightarrow \pi \left[\frac{n}{2} (2a + (n-1)d) \right]$$

$$S_{13} \rightarrow \pi \left[\frac{13}{2} (2(0.5) + (13-1)(0.5)) \right] \Rightarrow \frac{22}{7} \left[\frac{13}{2} \left(2 \times \frac{1}{2} + 12 \times \frac{1}{2} \right) \right] \Rightarrow \frac{22}{7} \left[\frac{13}{2} \times 7 \right] \Rightarrow 11 \times 13 \Rightarrow \boxed{143 \text{ cm}}$$



#LP: In a potato race, a bucket is placed at the starting point, which is 5m from the first potato, and the other potatoes are placed 3 m apart in a straight line. There are "n" potatoes in the line. Each competitor starts from the bucket, picks up the nearest potato, runs back with it, drops it in the bucket, runs back to pick up the next potato, runs to the bucket to drop it in the bucket, and she continues in the same way until all the potatoes are in the bucket. What is the total distance the competitor has to run?



distance
distance
"

$$\begin{aligned} \text{1st potato} &\Rightarrow 5 + 5 \Rightarrow 2 \times 5 \\ \text{2nd potato} &= 8 + 8 \Rightarrow 2 \times 8 \\ \text{3rd potato} &\Rightarrow 11 + 11 \Rightarrow 2 \times 11 \end{aligned}$$

$$\begin{aligned} \text{Total distance} &\Rightarrow 2 \times 5 + 2 \times 8 + 2 \times 11 + \dots \\ &\Rightarrow 2 [5 + 8 + 11 + \dots] \\ &\Rightarrow 2 \times S_n \end{aligned}$$

$$\begin{aligned} &\Rightarrow 2 \times \left[\frac{n}{2} (2(5) + (n-1)(3)) \right] \Rightarrow n [10 + 3n - 3] \\ &\Rightarrow n [7 + 3n] \end{aligned}$$

#LP: The school auditorium was to be constructed to accommodate at least 1500 people. The chairs are to be placed in concentric circular arrangement in such a way that each succeeding circular row has 10 seats more than the previous one.

$d = 10$ seat [CBSE SQP (Standard) 2022-23]

(i) If the first circular row has 30 seats, how many seats will be there in the 10th row?



AP: 30 seats, 40 seats, 50 seats - - -



$$a_{10} = a + 9d$$
$$= 30 + 9(10)$$

10th row $\Rightarrow$ 120 seats

K3B
1st H cheez ki AP k
ki jaroor mention krni

(ii) For 1500 seats in the auditorium, how many rows need to be there?

$$\begin{aligned}\text{Total seats} &= 1500 \\ S_n &= 1500 \\ \frac{n}{2} (2(30) + (n-1)(10)) &= 1500\end{aligned}$$

$$n = 15$$

Let 'n' rows are there.

if 1500 seats $\leftarrow$
then 15 rows.

(ii) If 1500 seats are to be arranged in the auditorium, how many seats are still left to be put after 10th row?

Total rows $\Rightarrow$ 15 rows

1st 2nd 3rd 10th 11th $\rightarrow$ 15th
30, 40, 50, ..., S₁₀

After 10th row, seats left $\Rightarrow$ total seats - seats from 1 to 10th row

$$1500 - S_{10}$$

$$1500 - \left[\frac{10}{2} (2(30) + (10-1)(10)) \right]$$

(iii) If there were 17 rows in the auditorium, how many seats will be there in the middle row?

$$\begin{aligned}n &= 17 \leftarrow \text{odd} \\m_7 &\Rightarrow \left(\frac{n+1}{2}\right)^{\text{th}} \\&= \left(\frac{17+1}{2}\right)^{\text{th}} \\&= \left(\frac{18}{2}\right)^{\text{th}} \rightarrow 9^{\text{th}}\end{aligned}$$

$$\begin{aligned}a_9 &= a + 8d \\&= 30 + 8(10) \\&= 30 + 80\end{aligned}$$

9^{th} row $\Rightarrow$ 110 seats

#LP: A school is organizing a charity run to raise funds for a local hospital. The run is planned as a series of rounds around a track. with each round being 300 metres. To make the event more challenging and engaging. the organizers decide to increase the distance of each subsequent round by 50 metres. For example, the second round will be 350 metres, the third round will be 400 metres and so on. The total number of rounds planned is 10.

1st round dist $\Rightarrow$ 300 m
2nd " " $\Rightarrow$ 350 m
3rd " " $\Rightarrow$ 400 m

Based on the information given above, answer the following questions:

i. Write the fourth, fifth, and sixth terms of the Arithmetic Progression so formed.

dist in each round $\rightarrow$ 300m, 350m, 400m, 450m, 500m, 550m
AP $\checkmark$
 $\uparrow$ 44
 $\uparrow$ 54
 $\uparrow$ 64

ii. Determine the distance of the 8th round.

$$a_8 = a + 7d$$
$$= 300 + 7(50)$$

↓
() m
An

iii. Find the total distance run after completing all 10 rounds.

$$S_{10} = \frac{10}{2} \left[\underline{\underline{2(300) + (10-1)(50)}} \right] \text{ m}$$

AP समाप्त!

but 2-3 Q left

THANK YOU COODIESS !!

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